



Towards Load Identification of Moving Sources from Distributed Strain Measurements on Composite Plates

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Abstract

This work addresses the problem of identifying moving dynamic loads acting on composite material structures, in cases where the excitation is not directly measurable but must be inferred from the structural response.

For this purpose, a case study consisting of a laminated composite panel with integrated optical Fiber Bragg Grating (FBG) sensors distributed on the surface is considered. These sensors are capable of providing time series of strain measurements. Such measurements represent the input to the reconstruction model and constitute the link between numerical simulation and identification of unknown quantities.

The work is divided into two main phases. In the first phase, a dynamic model of the panel is developed that integrates the description of the laminate, the sensor layout, and the formulation of the load associated with a moving source, characterized by an assigned spatio-temporal law. Modal analysis allows extraction of the dynamic properties of the structure independently from the load, identifying the relevant frequency band and the contribution of modes to the response. This information is used to coherently define the parameters of the transient simulation, such as time step, analysis duration, and sampling frequency, in order to accurately reproduce the dynamic response and the strain field at the sensor points.

In the second phase, the inverse problem is formulated, aimed at estimating the parameters of the load and the motion of the source (initial position, trajectory, and velocity) from strain measurements. The problem presents intrinsic critical issues, including sensitivity to noise, possible non-uniqueness of the solution, and dependence on sensor arrangement. These aspects are discussed in terms of reconstruction robustness, discriminative capacity of available signals, and physical consistency between hypothesized motion scenarios and observed response.

In summary, the work proposes an integrated chain of dynamic modeling and inverse identification for the estimation of moving loads on composite structures, with potential applications in structural monitoring and diagnostics of systems subjected to impulsive excitations in realistic operating conditions.

1 Project Requirements

To formulate the inverse problem, it is necessary to have a direct model capable of describing the dynamic response of the structure to the passage of the mobile source. This model provides the link between the load parameters and the strain measurements recorded by the sensors.

A laminated composite structure is therefore considered, modeled as a plate according to Kirchhoff-Love theory, with stack-up consistent with the real component and equipped with 8 optical sensors with Fiber Bragg Grating (FBG) distributed on the surface. The FBG sensors measure the local axial strain along the sensor axis, providing spatial discretization of the strain field.

In particular, the sensors arranged along the principal axes of the structure (FBG 1-4) are oriented according to the 0° and 90° directions, while the sensors positioned along the diagonals (FBG 5-8) are oriented at $\pm 45^\circ$. This configuration allows sampling of different components of the strain field and increases the sensitivity of the measurement system to different solicitation modes of the structure.

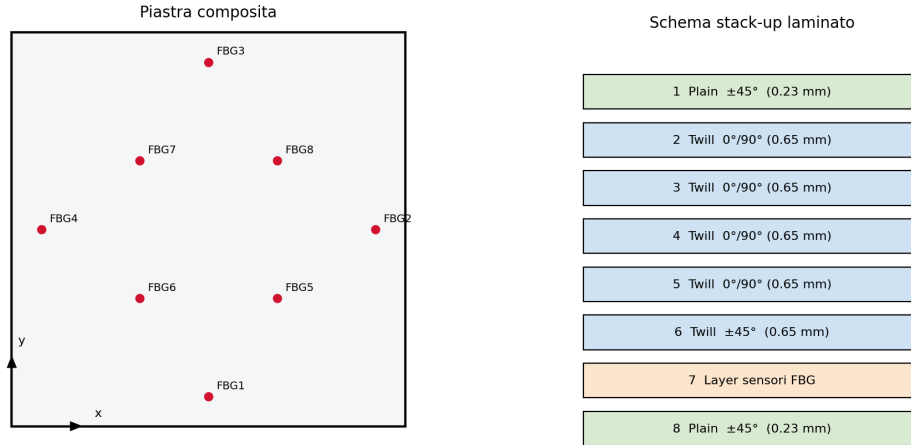


Figure 1: Schematic of the composite structure with FBG sensor arrangement and synthetic representation of the laminated stack-up.

The kinematics of the mobile source is described by

$$x_s(t) = (x_s)_0 + \dot{u} t, \quad y_s(t) = (y_s)_0 + \dot{v} t. \quad (1)$$

The source represents a propagating pressure wave. As a first approximation, it is modeled as a point source with cylindrical wavefront and geometric decay law inversely proportional to the distance from the source. Fluid-structure interaction is neglected.

The distance between the source and a point on the plate with coordinates x, y, z is therefore defined as

$$R(x, y, t) = \sqrt{(x - x_s(t))^2 + (y - y_s(t))^2 + z_s^2} \quad (2)$$

and the corresponding pressure field applied to the structure

$$p(x, y, t) = p_{\max} \frac{z_s}{R(x, y, t)} \sin^2 \left(\frac{\pi t}{T} \right). \quad (3)$$

where $z_s = 2\text{ m}$ represents the minimum distance between source and structure, $p_{\max}$ the peak value of the pressure, and T the duration of the impulse. The geometric decay law $1/R$ is consistent with the propagation of a point source in free field, used as a simplified model of the incident pressure wave.

On this basis, the dynamic model is constructed and used in the subsequent numerical analyses. In particular, a modal analysis and a transient simulation of the structural response are performed, maintaining behavior in the linear elastic regime.

Modal analysis allows identification of the relevant frequency band and the contribution of modes to the response, while transient analysis permits reconstruction of the time series of strain at the sensor points, which constitute the input data for the inverse problem.

The three kinematic scenarios reported in Table ?? have been selected to represent different excitation configurations: a centered passage, an eccentric passage, and motion along the diagonal direction.

Motion	$(x_s)_0$ [m]	$(y_s)_0$ [m]	$\dot{u}$ [m/s]	$\dot{v}$ [m/s]
1	-0.25	0.00	5	0
2	-0.25	1.00	5	0
3	-0.50	-0.50	10	10

Table 1: Kinematic characteristics of the three load motions studied.

2 Analytical Model

2.1 Objective and Problem Definition

The following analytical model aims to provide a simplified theoretical representation capable of supporting the interpretation of the results from the subsequent FEM numerical analysis.

The structure under examination can be modeled as a thin plate subjected to a distributed dynamic load $q(x, y, t)$. According to the Kirchhoff-Love plate theory, the dynamic behavior of the plate is described by the following partial differential equation:

$$D_{11} \frac{\partial^4 w}{\partial x^4} + 4D_{16} \frac{\partial^4 w}{\partial x^3 \partial y} + 2(D_{12} + 2D_{66}) \frac{\partial^4 w}{\partial x^2 \partial y^2} + 4D_{26} \frac{\partial^4 w}{\partial x \partial y^3} + D_{22} \frac{\partial^4 w}{\partial y^4} + \rho h \ddot{w}(x, y, t) = q(x, y, t) \quad (4)$$

In Eq. (4):

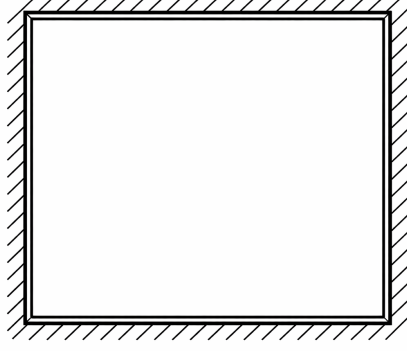
- $w(x, y, t)$ represents the transverse displacement of the plate;
- ρ is the material density;
- h is the plate thickness;
- D_{ij} are the coefficients of the flexural stiffness matrix of the laminate;
- $q(x, y, t)$ represents the distributed load acting on the surface.

Since Eq. (4) is a partial differential equation in the variable $w(x, y, t)$, its solution requires the definition of spatial boundary conditions and temporal initial conditions.

In the real component, the composite panel is part of a larger structure and is constrained along the perimeter through mechanical connections with support elements. Since the interest of this study is focused on the dynamic response of the central sensorized zone, a *Region of Interest* (ROI) is introduced, defined as the portion of structure where the FBG sensors are located and where the deformative response is analyzed.

The ROI is therefore modeled as a square plate of side L , assuming that the constraint exerted by the surrounding structure can be approximated with fixed support boundary conditions along the edge Γ . This assumption allows reduction of model complexity while maintaining a sufficiently accurate representation of local dynamic response. Additionally, the fixed support assumption represents a conservative approximation of the stiffness of the surrounding structure and allows isolation of the local dynamic response in the sensorized region.

The boundary and initial conditions thus assume the form:



$$\begin{cases} w(-L/2, y, t) = w(L/2, y, t) = 0 \\ w(x, -L/2, t) = w(x, L/2, t) = 0 \\ \frac{\partial w(-L/2, y, t)}{\partial x} = \frac{\partial w(L/2, y, t)}{\partial x} = 0 \\ \frac{\partial w(x, -L/2, t)}{\partial y} = \frac{\partial w(x, L/2, t)}{\partial y} = 0 \\ w(x, y, 0) = 0 \\ \frac{\partial w(x, y, 0)}{\partial t} = 0 \end{cases}$$

The origin of the reference system is placed at the geometric center of the plate, with axes aligned to the sides of the structure.

2.2 Classical Laminate Theory

Since the plate is made of composite material, the flexural stiffness appearing in Eq. (??) can be derived using the *Classical Laminate Theory* (CLT).

For each lamina k , the reduced stiffness matrix is defined in the material system (1, 2)

$$\mathbf{Q}^{(k)} = \begin{bmatrix} Q_{11}^{(k)} & Q_{12}^{(k)} & 0 \\ Q_{12}^{(k)} & Q_{22}^{(k)} & 0 \\ 0 & 0 & Q_{66}^{(k)} \end{bmatrix}.$$

This matrix is transformed in the global system (x, y) as a function of the orientation angle θ_k , obtaining $\bar{\mathbf{Q}}^{(k)}$.

Denoting with $z_0, \dots, z_N$ the coordinates of the lamination interfaces with respect to the mid-plane, the coefficients of the flexural stiffness matrix are given by

$$D_{ij} = \frac{1}{3} \sum_{k=1}^N \bar{Q}_{ij}^{(k)} (z_k^3 - z_{k-1}^3), \quad i, j \in \{1, 2, 6\}.$$

Thus, the anisotropic flexural stiffness matrix is obtained:

$$\mathbf{D} = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix},$$

which directly enters the Kirchhoff-Love formulation for anisotropic laminated plates.

2.3 Energy Formulation

Given the complexity of the differential equation Eq. (??), an energy approach based on Hamilton's variational principle is adopted:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{q}} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \right) = 0 \quad (5)$$

where the Lagrangian is defined as

$$\mathcal{L} = T - U - V$$

$$T = \frac{1}{2} \int_{\Omega} \rho h \left(\frac{\partial w}{\partial t} \right)^2 dx dy$$

$$U = \frac{1}{2} \int_{\Omega} (D_{11} w_{xx}^2 + 2D_{12} w_{xx} w_{yy} + D_{22} w_{yy}^2 + 4D_{66} w_{xy}^2 + 4D_{16} w_{xx} w_{xy} + 4D_{26} w_{yy} w_{xy}) dx dy$$

$$V = - \int_{\Omega} q(x, y, t) w(x, y, t) dx dy$$

In the calculation of kinetic energy, it is assumed that in-plane velocities are negligible, i.e., $\dot{u} \simeq 0$ and $\dot{v} \simeq 0$.

Using a Rayleigh-Ritz approach, the solution is assumed in the form

$$w(x, y, t) = \sum_{i=0}^n \sum_{j=0}^m q_{ij}(t) \phi_{ij}(x, y)$$

$$\phi_{ij}(x, y) = \left(x^2 - \frac{L^2}{4} \right)^2 \left(y^2 - \frac{L^2}{4} \right)^2 x^i y^j$$

The symmetries of the problem in the three cases studied additionally allow simplification of the summations:

1. The motion of the source is parallel to the x axis and passing through the origin: the deformation is symmetric with respect to the xz plane, therefore the ϕ_{ij} with odd j are zero.
2. The motion is eccentric and with generic direction: it is not possible to impose symmetries a priori.
3. The motion is symmetric with respect to the diagonal: we have $\phi_{ij} = \phi_{ji}$.

By imposing stationarity of the action, a system of ordinary differential equations in the generalized coordinates $q_{ij}(t)$ is finally obtained, whose solution allows the transverse displacement $w(x, y, t)$ to be obtained.

2.4 Determination of Generalized System Matrices

In the reduced modal domain, the free dynamics of the plate is described by

$$\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{0}.$$

The generalized matrices are obtained by projecting the inertial and flexural contributions onto the Ritz basis:

$$M_{pq} = \rho h \int_{\Omega} \phi_p \phi_q d\Omega,$$

$$K_{pq} = \int_{\Omega} \left[D_{11} \phi_{p,xx} \phi_{q,xx} + D_{22} \phi_{p,yy} \phi_{q,yy} + D_{12} (\phi_{p,xx} \phi_{q,yy} + \phi_{p,yy} \phi_{q,xx}) \right. \\ \left. + 4D_{66} \phi_{p,xy} \phi_{q,xy} + 2D_{16} (\phi_{p,xx} \phi_{q,xy} + \phi_{p,xy} \phi_{q,xx}) + 2D_{26} (\phi_{p,yy} \phi_{q,xy} + \phi_{p,xy} \phi_{q,yy}) \right] d\Omega.$$

Assuming a harmonic response in time,

$$\mathbf{q}(t) = \hat{\mathbf{q}} e^{i\omega t},$$

we have

$$\ddot{\mathbf{q}}(t) = -\omega^2 \hat{\mathbf{q}} e^{i\omega t},$$

and substituting into the free equation gives

$$(\mathbf{K} - \omega^2 \mathbf{M}) \hat{\mathbf{q}} = \mathbf{0}.$$

The condition for the existence of non-trivial solutions leads to the generalized eigenvalue problem

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0 \iff \mathbf{K} \hat{\mathbf{q}} = \lambda \mathbf{M} \hat{\mathbf{q}}, \lambda = \omega^2.$$

From the eigenvalues λ_k the natural angular frequencies $\omega_k = \sqrt{\lambda_k}$ and finally the natural frequencies are obtained

$$f_k = \frac{\omega_k}{2\pi}.$$

2.5 Natural Frequencies of the Analytical Model

Execution of the developed code provides the natural frequencies through Rayleigh-Ritz modal analysis and relative convergence test as $m = n$ increases.

m	f_1 [Hz]	f_2 [Hz]	f_3 [Hz]	f_4 [Hz]	f_5 [Hz]	f_6 [Hz]
4	50.91	102.97	102.97	158.97	187.17	188.32
6	50.90	102.90	102.90	158.87	180.54	181.49
8	50.90	102.90	102.90	158.87	180.29	181.23
10	50.90	102.90	102.90	158.87	180.28	181.22

Table 2: Convergence of the first six natural frequencies obtained with the Rayleigh-Ritz model.

The results show rapid convergence already for $m \geq 8$.

3 Model Preparation

3.1 Geometry Modeling

This section describes the modeling of the real structure within a Finite Element analysis framework. In Fig. ?? the geometry of the component under study is shown.

Given the complexity of the initial CAD model, appropriate simplification operations were performed using *SpaceClaim* and *SolidWorks*, with the objective of reducing computational cost and simulation time. In particular, grooved sections and all geometric details not relevant to the global structural response were removed.

Since the panel is made of composite material, it is convenient to adopt modeling using *shell* elements, assuming plane stress state validity. Consequently, the solid model was replaced by only the external surface, namely the portion of structure directly subjected to the action of the pressure wave.

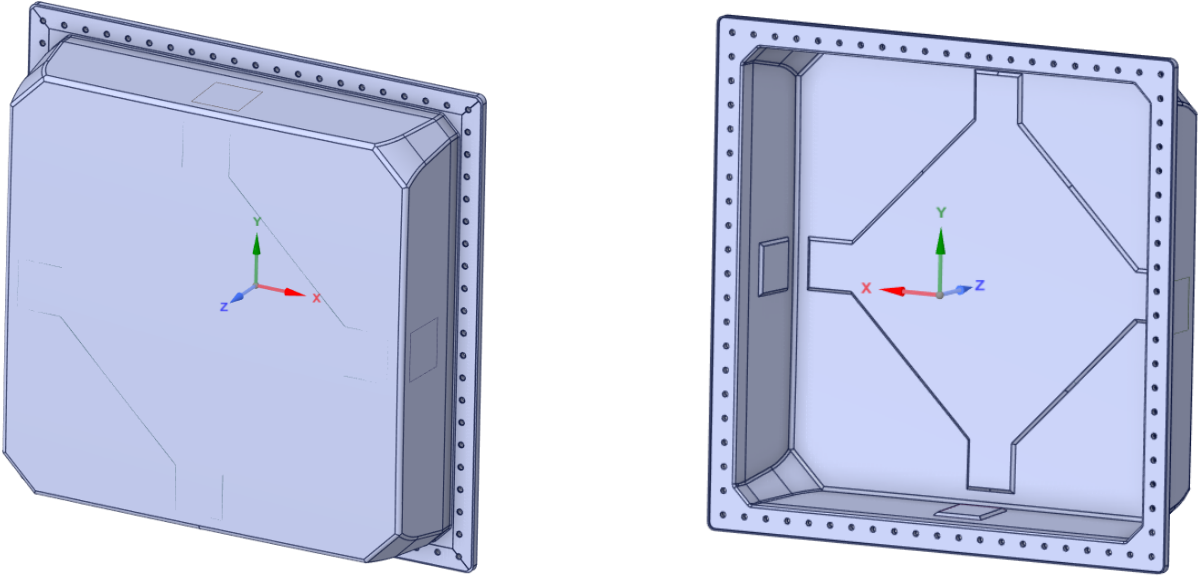


Figure 2: Geometry before simplification

In order to obtain an accurate estimate of the *strain* at the FBG sensor locations, it was necessary to guarantee the presence of nodes exactly at the measurement positions. An initial strategy consisted of using a structured mesh; however, the presence of chamfers at the plate corners made this approach impractical.

An alternative strategy based on the introduction of *hardpoints* in the geometry using ANSYS APDL was therefore adopted. *Hardpoints* are *keypoints* inserted a posteriori on the geometry, to which specific coordinates are assigned and which allow forcing the presence of nodes at desired positions.

To correctly apply this procedure, it was necessary to make the ROI actually planar; a preliminary verification in *SolidWorks* had indeed revealed a slight non-planarity of the surface. A script was therefore developed that imports the geometry (Parasolid format),

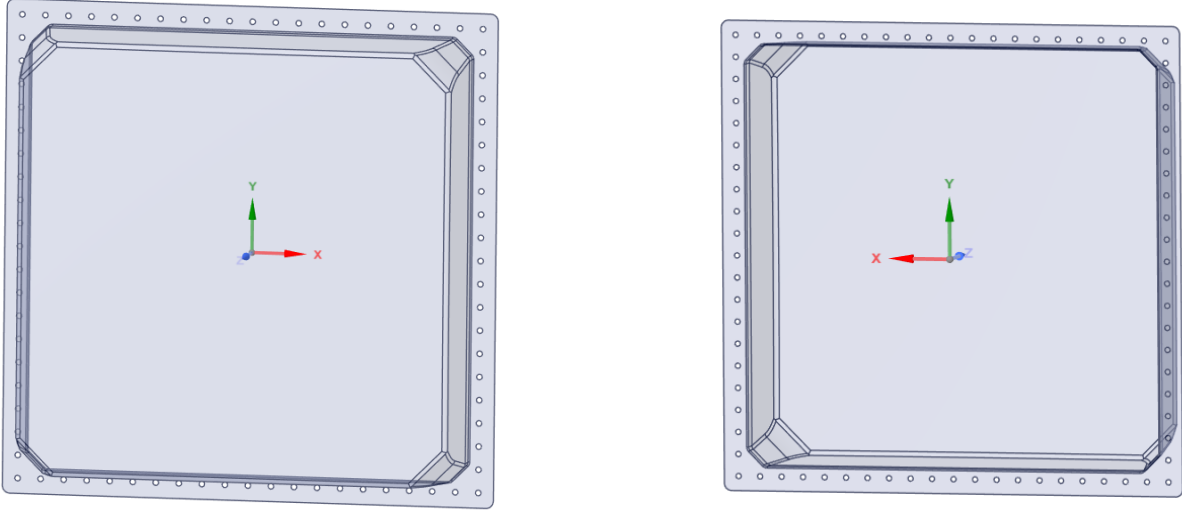


Figure 3: Simplified geometry (external and internal surfaces)

identifies the ROI node with maximum z coordinate, and realigns all other ROI nodes to that elevation.

During this phase, problems emerged in mesh generation, particularly near the corners, due to the presence of faces with more than four edges and characterized by high curvature. These critical issues were resolved by intervening on the CAD model in *SolidWorks*, through the removal of problematic surfaces and their reconstruction using *boundary surface* and *filled surface*.

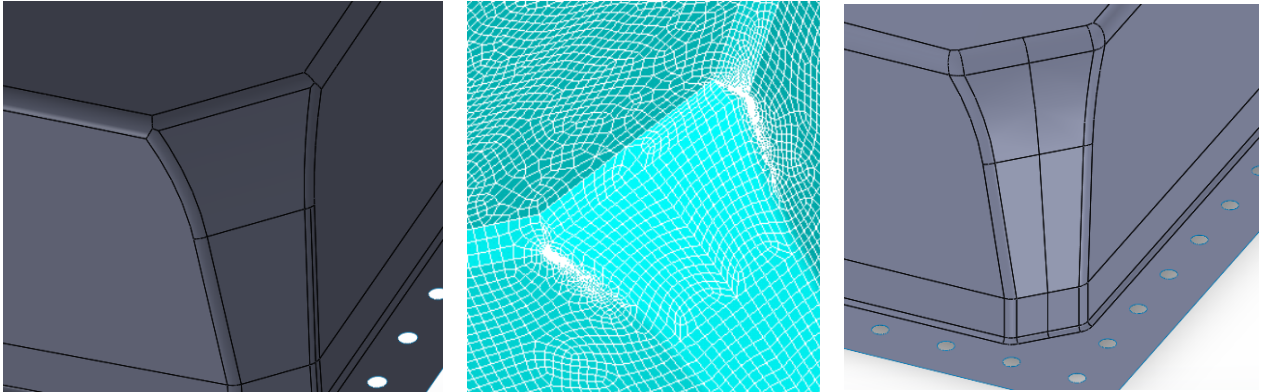


Figure 4: Problems encountered in modeling geometric details and resolution after cleanup

3.2 Material Definition

Materials were initially defined in *Workbench* and subsequently implemented in the APDL environment, in order to guarantee greater flexibility in model management.

Properties of Outline Row 3: Plain Fiber				
	A	B	C	
1	Property	Value	Unit	
2	Material Field variables	Table		
3	Density	1400	kg m ⁻³	
4	Orthotropic Elasticity			
5	Young's Modulus X direction	53	GPa	
6	Young's Modulus Y direction	53	GPa	
7	Young's Modulus Z direction	1	GPa	
8	Poisson's Ratio XY	0.2		
9	Poisson's Ratio YZ	0.3		
10	Poisson's Ratio XZ	0.3		
11	Shear Modulus XY	3.3	GPa	
12	Shear Modulus YZ	1	GPa	
13	Shear Modulus XZ	1	GPa	
14	Ply Type			
15	Type	Woven		

Properties of Outline Row 3: Twill Fiber				
	A	B	C	
1	Property	Value	Unit	
2	Material Field variables	Table		
3	Density	1400	kg m ⁻³	
4	Orthotropic Elasticity			
5	Young's Modulus X direction	59	GPa	
6	Young's Modulus Y direction	59	GPa	
7	Young's Modulus Z direction	1	GPa	
8	Poisson's Ratio XY	0.2		
9	Poisson's Ratio YZ	0.3		
10	Poisson's Ratio XZ	0.3		
11	Shear Modulus XY	3.3	GPa	
12	Shear Modulus YZ	1	GPa	
13	Shear Modulus XZ	1	GPa	
14	Ply Type			
15	Type	Woven		

Figure 5: Fiber properties (Workbench Mechanical)

3.3 Meshing and Laminate Creation

As anticipated, the modeling of composites was performed using *shell* elements, specifically **SHELL281**, preferred over **SHELL181** for greater accuracy in strain field reconstruction thanks to the higher number of nodes.

Some default element settings were modified: $\text{KEYOPT}(5) = 1$ to neglect initial curvature effects (the ROI is assumed planar) and $\text{KEYOPT}(8) = 1$ to save *strain* on both faces of each layer.

Three integration points along the thickness were adopted for most layers, increased to five for those adjacent to FBG sensors, in order to improve local resolution of the deformation field.

The initial element size was set to 0.05 m, subsequently refined through a convergence analysis until obtaining negligible variations in the results.

To ensure symmetry and comparability between sensor measurements, one quarter of the plate was meshed, subsequently applying symmetry operations (NSYM, ESYM) and node/element merging.

The definition of the stack-up was performed using commands from the *SEC* family.

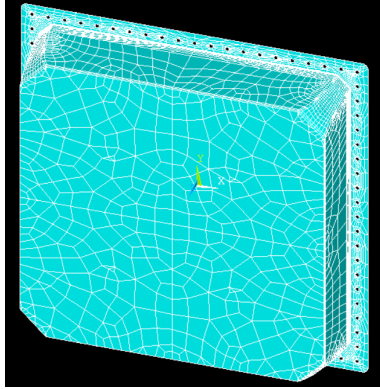


Figure 6: Mesh obtained with sizing 0.005 meters

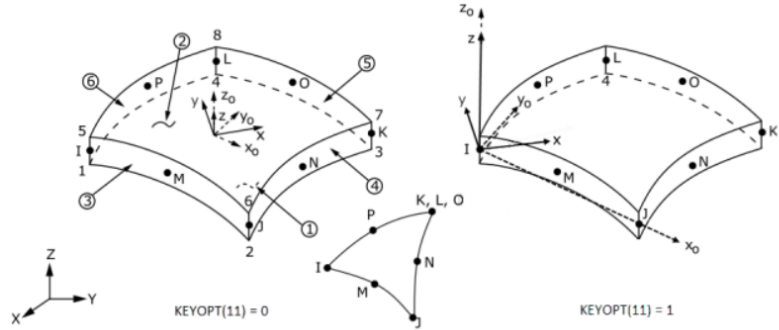


Figure 7: SHELL281 element

Laminate modeling in APDL furthermore requires explicit management of local element orientation. For this purpose, a script was developed that, through Gram-Schmidt orthogonalization and use of Euler angles, constructs a local frame for each element based on its normal.

This procedure ensures coherence in fiber orientation, additionally imposing that the dot product between global x axis and local axis is non-negative. The algorithm is computationally expensive ($\mathcal{O}(N)$), but can be optimized by grouping elements with similar orientation.

3.4 Imposition of Constraints

The panel is constrained to the surrounding structure through $\Phi 12$ holes distributed along the edge. Rigorous modeling would require verification of bolts against sliding and pull-out; however, assuming the use of bolts of at least M8 class, it is reasonable to model the connection as a rigid constraint (*fixed support*).

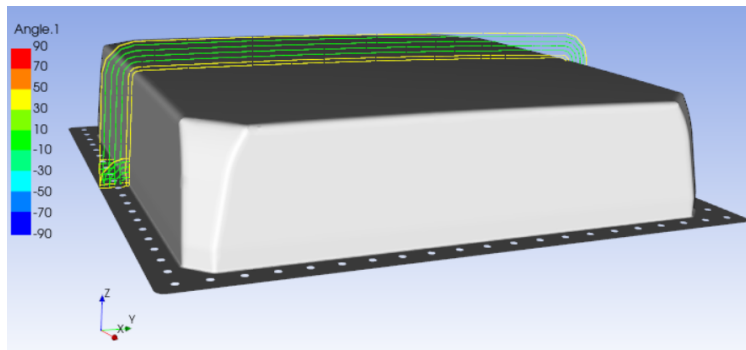


Figure 8: Skin angulation (ACP Pre)

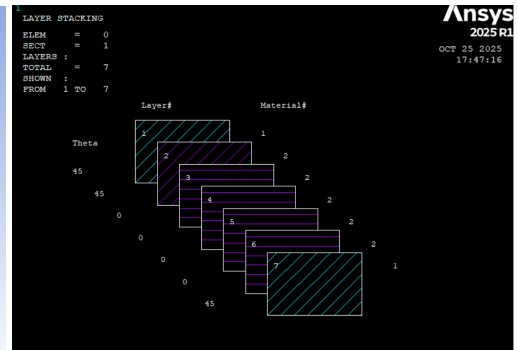


Figure 9: Stackup definition

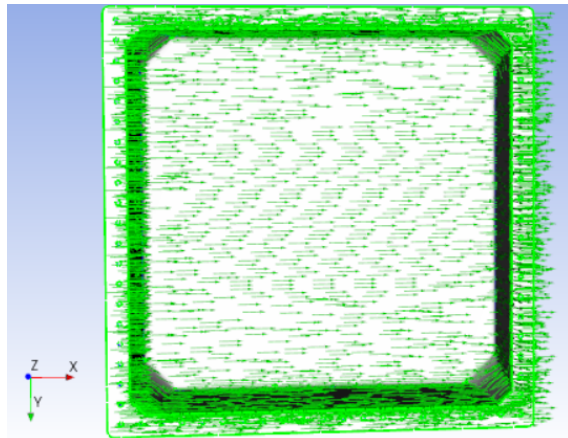


Figure 10: Fiber direction (ACP Pre)

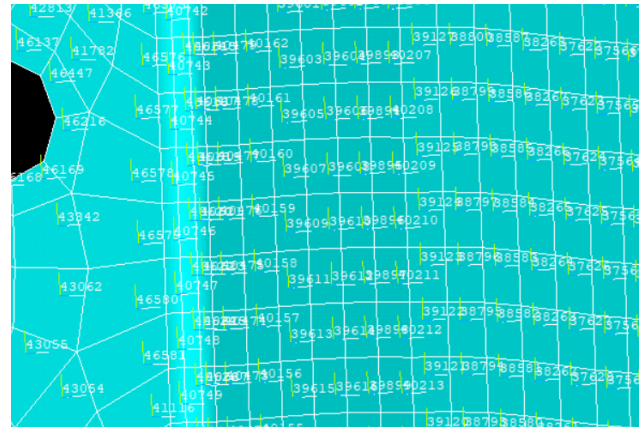


Figure 11: Local reference systems

The preload distribution plates were not included in the model, as they are not directly relevant to the analyzed dynamic response.

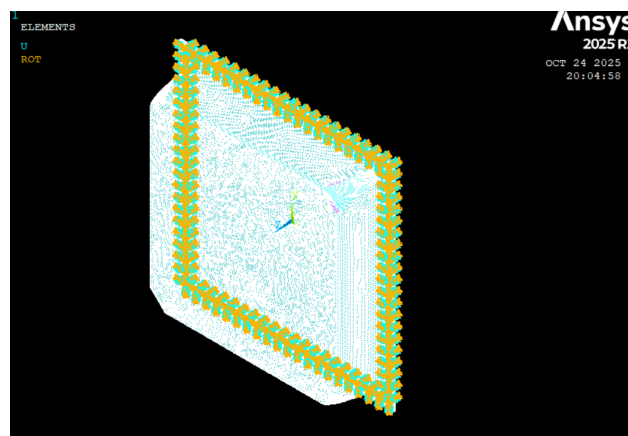


Figure 12: Imposed constraints

4 Modal Analysis

In order to characterize the dynamic behavior of the structure and identify the relevant frequencies with respect to the imposed excitation, a modal analysis was conducted.

Modal analysis is a fundamental tool for dynamic characterization of structures and identification of natural vibration modes [?]:

$$M\ddot{\mathbf{D}}(t) + K\mathbf{D}(t) = 0$$

Assuming a harmonic solution in time, $\mathbf{D}(t) = \Phi e^{i\omega t}$, the problem is reduced to eigenvalue form:

$$M^{-1}K\Phi = \lambda\Phi$$

from which the natural angular frequencies $\omega = \sqrt{\lambda}$ and the corresponding natural modes Φ are obtained.

Since the structure is symmetric and no damping was introduced, the *Block Lanczos* solver was used, adopting the constraints defined during the modeling phase.

Not knowing a priori the modes relevant in the frequency band of interest, the analysis was extended to the interval $[0, 300]$ Hz, equal to approximately 30 times the characteristic frequency of the load. This choice ensures the inclusion of all potentially excited modes.

The characteristic frequency of the load can be derived from its temporal expression:

$$p(x, y, t) = k \sin^2 \left(\frac{\pi t}{T} \right) = \frac{k}{2} \left(1 - \cos \left(\frac{2\pi t}{T} \right) \right)$$

from which a principal harmonic component with frequency is identified as

$$f_{ecc} = \frac{1}{T} = 10 \text{ Hz.}$$

To assess the actual contribution of individual modes to the dynamic response, the modal participation factor was analyzed. This parameter provides a measure of the energy associated with each mode in the direction of interest.

In the case examined, the load acts predominantly in the transverse direction (z), therefore only participation factors along this direction were considered.

From the joint analysis of natural frequencies and participation factors, it emerges that the dynamic contribution is dominated by the fundamental mode, while higher modes are poorly excited.

Consequently, the dynamically relevant band can be restricted to $[0, 100]$ Hz and the structure operates in a sub-resonant regime, as evidenced by the ratio:

$$\frac{f_{ecc}}{f_1} \simeq 0.196$$

This result indicates that the excitation acts at frequencies significantly lower than the first natural frequency, with a response dominated by the quasi-static behavior of the system.

FROM	MODE	1	ITEM1=PFAC	ITEM2=DIRE Z	VALUE= 1.18506810
FROM	MODE	2	ITEM1=PFAC	ITEM2=DIRE Z	VALUE= 0.932144570E-03
FROM	MODE	3	ITEM1=PFAC	ITEM2=DIRE Z	VALUE= 0.729132521E-04
FROM	MODE	4	ITEM1=PFAC	ITEM2=DIRE Z	VALUE=-0.893423059E-03
FROM	MODE	5	ITEM1=PFAC	ITEM2=DIRE Z	VALUE= 0.216301463E-03
FROM	MODE	6	ITEM1=PFAC	ITEM2=DIRE Z	VALUE= 0.670184234
FROM	MODE	7	ITEM1=PFAC	ITEM2=DIRE Z	VALUE=-0.490852853E-03
FROM	MODE	8	ITEM1=PFAC	ITEM2=DIRE Z	VALUE=-0.144987044E-02

Figure 13: Modal participation factors

SET	TIME/FREQ	LOAD STEP	SUBSTEP	CUMULATIVE
1	50.894248	1	1	1
2	105.77069	1	2	2
3	105.81965	1	3	3
4	159.31837	1	4	4
5	190.74690	1	5	5
6	192.24875	1	6	6
7	244.24769	1	7	7
8	245.34032	1	8	8

Figure 14: Natural frequencies of the structure

In the following figure are shown the deformations associated with the fundamental mode and, for comparison, the sixth natural mode.

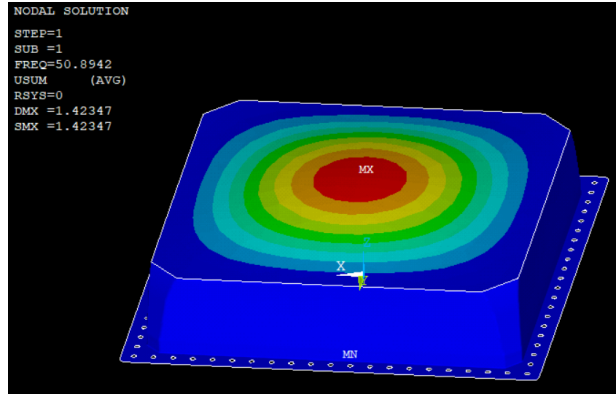


Figure 15: First natural mode

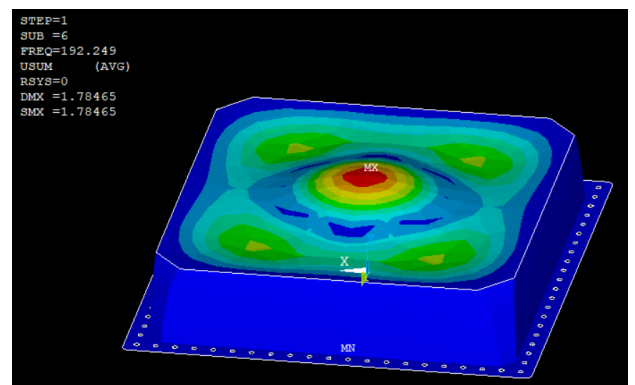


Figure 16: Sixth natural mode

5 Transient Analysis

The dynamic analysis of the structure in the presence of time-varying loads is governed by the equation of motion of the discrete system:

$$M\ddot{\mathbf{D}}(t) + C\dot{\mathbf{D}}(t) + K\mathbf{D}(t) = \mathbf{F}(t)$$

where $\mathbf{D}(t)$ represents the vector of nodal displacements, M the mass matrix, C the damping matrix, K the stiffness matrix, and $\mathbf{F}(t)$ the load vector.

The native APDL solvers allow solving this problem using two main approaches: modal superposition and direct numerical integration.

The modal superposition method is particularly efficient as it allows decoupling the system of equations in modal coordinates. However, in the case examined, the presence of a load varying both in space and time makes this approach less convenient, as it would require recalculation of generalized forces at each time step.

A direct time integration approach was therefore chosen, using the HHT- α method (Hilber-Hughes-Taylor), a generalization of the Newmark method.

This method introduces an α parameter that allows control of numerical damping, acting as a low-pass filter: high-frequency components are damped, while the frequency band of interest remains substantially unchanged.

In the absence of physical damping and assuming linear behavior of the structure, the term C can be neglected, simplifying the equation of motion.

The integration parameters were initially chosen based on the indications provided by ANSYS documentation, with the objective of capturing both the load application phase and the subsequent free transient.

Element size	0.05 m
α	0
Δt	10^{-3} s
T_{tot}	0.25 s
f_{camp}	250 Hz

Table 3: Initial simulation parameters

The parameter α regulates the amount of numerical damping, allowing attenuation of high-frequency components that are not relevant to the phenomenon being examined. The sampling frequency is directly related to the time step and will subsequently be adapted within the convergence analysis.

The choice of temporal parameters is consistent with the results of modal analysis: since the response is dominated by the first mode and characterized by relatively low frequencies, a sufficiently large time step can be adopted without compromising solution accuracy.

6 Convergence Analysis

The problem presents three main degrees of freedom from the numerical perspective: mesh dimension, time integration step, and magnitude of numerical damping. To avoid interference between these parameters, their influence was analyzed separately through progressive convergence study. A percentage variation lower than 0.5% was adopted as convergence criterion.

6.1 Spatial Convergence

The spatial convergence of the FEM model was verified by comparing results obtained with the initial mesh (*sizing* 0.05 m) with those from progressively finer meshes.

In particular, mesh refinement up to *sizing* of 0.025 m was considered, repeating modal analysis under the same conditions.

SET	TIME/FREQ	LOAD STEP	SUBSTEP	CUMULATIVE
1	50.840207	1	1	1
2	105.59117	1	2	2
3	105.92975	1	3	3
4	158.94414	1	4	4
5	190.95699	1	5	5
6	191.88353	1	6	6
7	244.03348	1	7	7
8	244.57890	1	8	8

ε_1	0.106%
ε_2	0.170%
ε_3	-0.104%
ε_4	0.235%
ε_5	-0.110%
ε_6	0.190%
ε_7	0.088%
ε_8	0.311%

Table 4: Percentage variations of frequencies

An average variation lower than 0.112% is observed, well below the preset convergence threshold. Therefore, further mesh refinement in the ROI is not necessary.

6.2 Temporal Convergence

Temporal convergence was evaluated by analyzing the energy of the strain signal associated with the FBG sensors, calculated in L^2 norm.

$$E_s = \int_0^T s(t)^2 dt \quad \Rightarrow \quad E_s = \sum_{k=0}^N s_k^2 \Delta t$$

Taking as reference the result obtained with the smallest timestep, the following indicator was introduced:

$$\eta_i(\Delta t) = \left| \frac{E_s(\Delta t) - E_s(\Delta t')}{E_s(\Delta t')} \right|_i$$

$$\max_i \eta_i(\Delta t^*) < 0.5\%$$

where i denotes the FBG sensor and $E_s(\Delta t)$ represents the energy calculated with time step Δt .

Simulations were therefore performed for 25 ms with timesteps equal to:

$$1\text{ ms}, 0.5\text{ ms}, 0.25\text{ ms}, 0.125\text{ ms}, 0.0625\text{ ms}$$

obtaining the results reported in the following tables.

The results show that the timestep necessary to satisfy the convergence criterion must be less than 0.125 ms . Therefore, a time step of 0.0625 ms was adopted.

For completeness, an additional convergence verification was performed using the finer mesh and comparing timesteps equal to:

$$1\text{ ms}, 0.5\text{ ms}, 0.25\text{ ms}, 0.125\text{ ms}$$

The computational cost of the simulation is proportional to the number of degrees of freedom of the system and the number of time steps:

$$C_{tot} \propto \frac{DOFs \cdot T_{tot}}{\Delta t}$$

Therefore, a mesh with *sizing* of 0.025 m and timestep of 0.5 ms was adopted.

The mesh refinement is limited to the ROI region, keeping the total number of system nodes nearly unchanged (passing from 34760 to 35430).

$$C_1 \propto \frac{6 \cdot 34760 \cdot 0.150}{0.0000625}$$

$$C_2 \propto \frac{6 \cdot 35430 \cdot 0.150}{0.0005}$$

Use of the coarse mesh with reduced timestep would therefore be approximately eight times more computationally expensive.

The load distribution was not considered in determining the timestep since pressure is applied simultaneously to all ROI nodes. Under such conditions, no local discontinuities of the pressure front occur.

Furthermore, the parameter governing the pressure gradient is dominated by the term R at each timestep:

$$\sqrt{R^2 + (x - \dot{u}t)^2 + (y - \dot{v}t)^2} \simeq R$$

6.3 Numerical Damping

The numerical damping coefficient was estimated by analyzing the evolution of the modal coordinate associated with the first mode during a free vibration simulation.

To obtain a sufficient number of oscillations on which to calculate the logarithmic decrement, the total simulation time was set to 100 ms .

The modal coordinate was obtained by projecting node displacements corresponding to the FBG sensors onto the first natural mode:

$$q_i = \frac{\phi_1^T u(t_i)}{\phi_1^T \phi_1}$$

The following table reports the evolution of the first five peaks of free oscillation for $\alpha = 0.05$.

$$\zeta = \frac{1}{8\pi} \log \left(\frac{A_1}{A_5} \right) \simeq 0.11\%$$

7 Discussion of Results

After defining the simulation parameters, three transient analyses corresponding to different loading conditions were performed. Below are the strain curves, expressed in microstrain, relative to the first four and last four FBG sensors.

The determination of the applied load, both in terms of eccentricity and direction of propagation, is difficult to address without a priori information. However, comparative analysis of the measurements provided by the sensors allows identification of characteristic patterns useful for qualitative reconstruction of the source motion. Problems of this type fall within the field of dynamic structure identification, in which load quantities or system parameters are estimated from response measurements [?].

The operating principle of FBGs relates axial deformation along the sensor with variation of the refractive index. The available measurements can therefore be interpreted as:

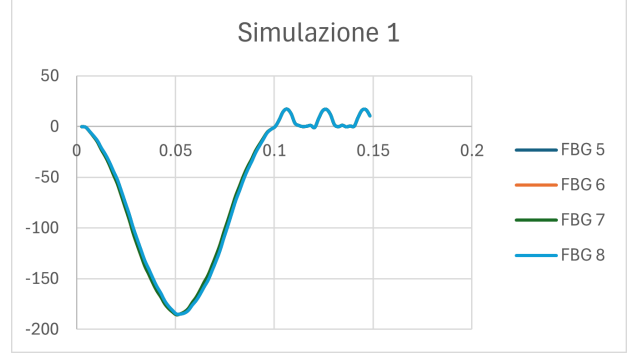
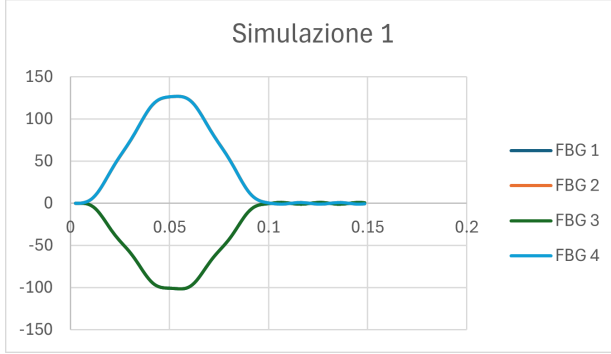
$$\begin{cases} \text{FBG 1, FBG 3} \rightarrow \varepsilon_x \\ \text{FBG 2, FBG 4} \rightarrow \varepsilon_y \\ \text{FBG 5, FBG 6, FBG 7, FBG 8} \rightarrow \varepsilon_{45} = \frac{\varepsilon_x + \varepsilon_y}{2} + \left| \frac{\gamma_{xy}}{2} \right| \end{cases}$$

where the absolute value of the shear term is related to sensor orientation along the diagonals.

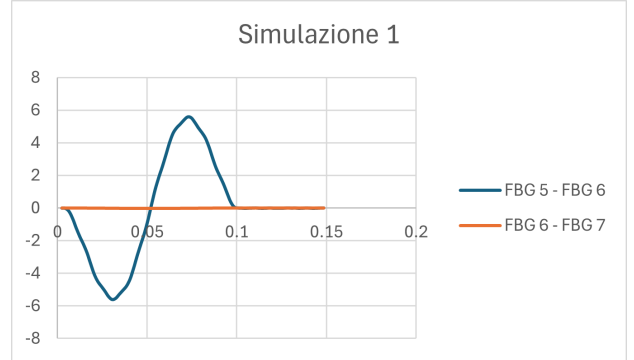
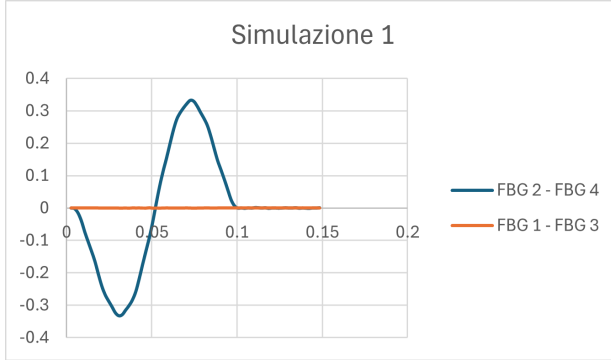
Individual strain curves are not, by themselves, sufficient to characterize the problem. For this reason, among the results considered are also the differences between sensors that are symmetric and antisymmetric with respect to the geometry.

Simulation 1

The load translates along the x axis with velocity $u_s = 0.5 \text{ m/s}$ and initial position $(-0.25, 0) \text{ m}$. The problem thus presents symmetry with respect to the xz plane.



The strain curves confirm this symmetry: sensors symmetric with respect to the xz plane show overlapping behaviors, while those with respect to the yz plane show no significant differences.

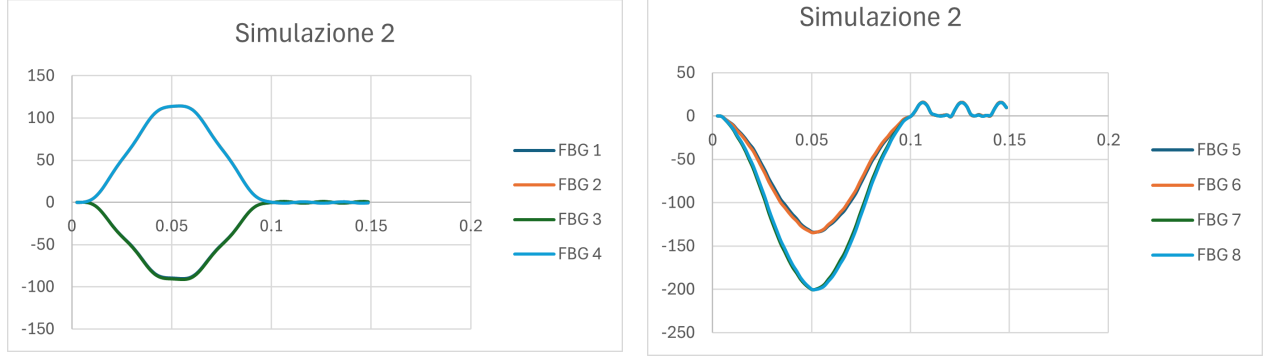


Analysis of the differences between sensors clearly highlights the source motion: until $t = 0.05 \text{ s}$ the strain is greater in sensors with smaller x coordinate, while subsequently the behavior reverses, indicating the passage of the source through the center of the plate.

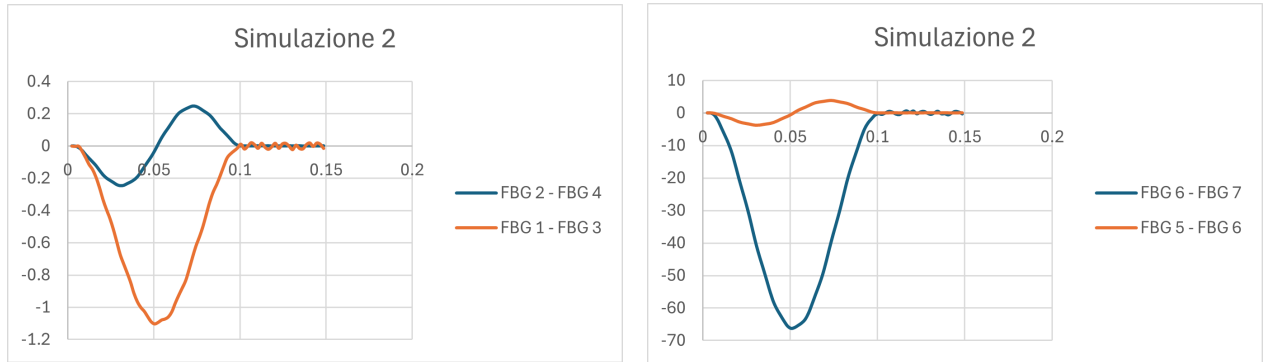
The negligible difference between sensors symmetric with respect to xz confirms the absence of motion components and eccentricity along the y direction.

Simulation 2

The load translates along a line parallel to the x axis, with velocity $u_s = 0.5 \text{ m/s}$ and initial position $(-0.25, 1) \text{ m}$. In this case there is no plane of symmetry.



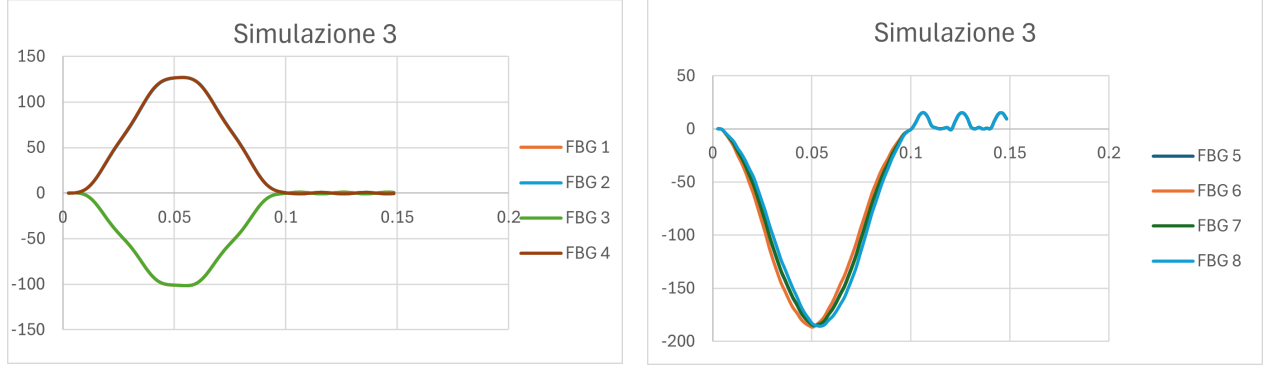
The curves show clear asymmetry along the y direction: sensors 7 and 8 record higher strain values compared to sensors 5 and 6, indicating the presence of eccentricity.



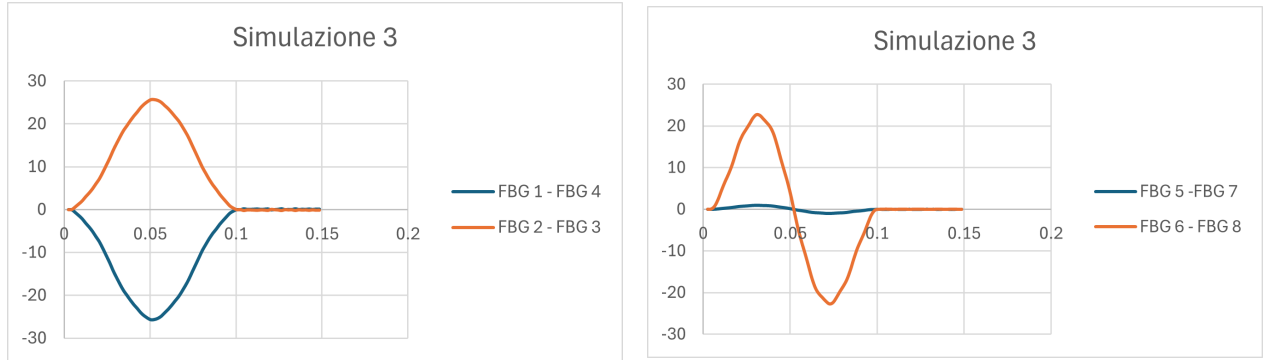
The differences between sensors confirm both the presence of eccentricity and a direction of propagation parallel to the x axis, similar to the previous case. In this case too, we observe the reversal of deformations corresponding to the passage of the source.

Simulation 3

The load moves along the line $y = x$ with velocity $\mathbf{v} = (10, 10) m/s$ and initial position $(-0.5, -0.5) m$. The plane of symmetry contains the direction of motion and the z axis.



The strain curves do not allow immediate identification of the motion. However, a clear difference is observed between the diagonal sensors: 5 and 7 present similar behaviors, while 6 and 8 are different.



Analysis of the differences allows identification of the direction of load propagation: the relationships between pairs 6-8 highlight simultaneously effects of motion and eccentricity along the diagonal.

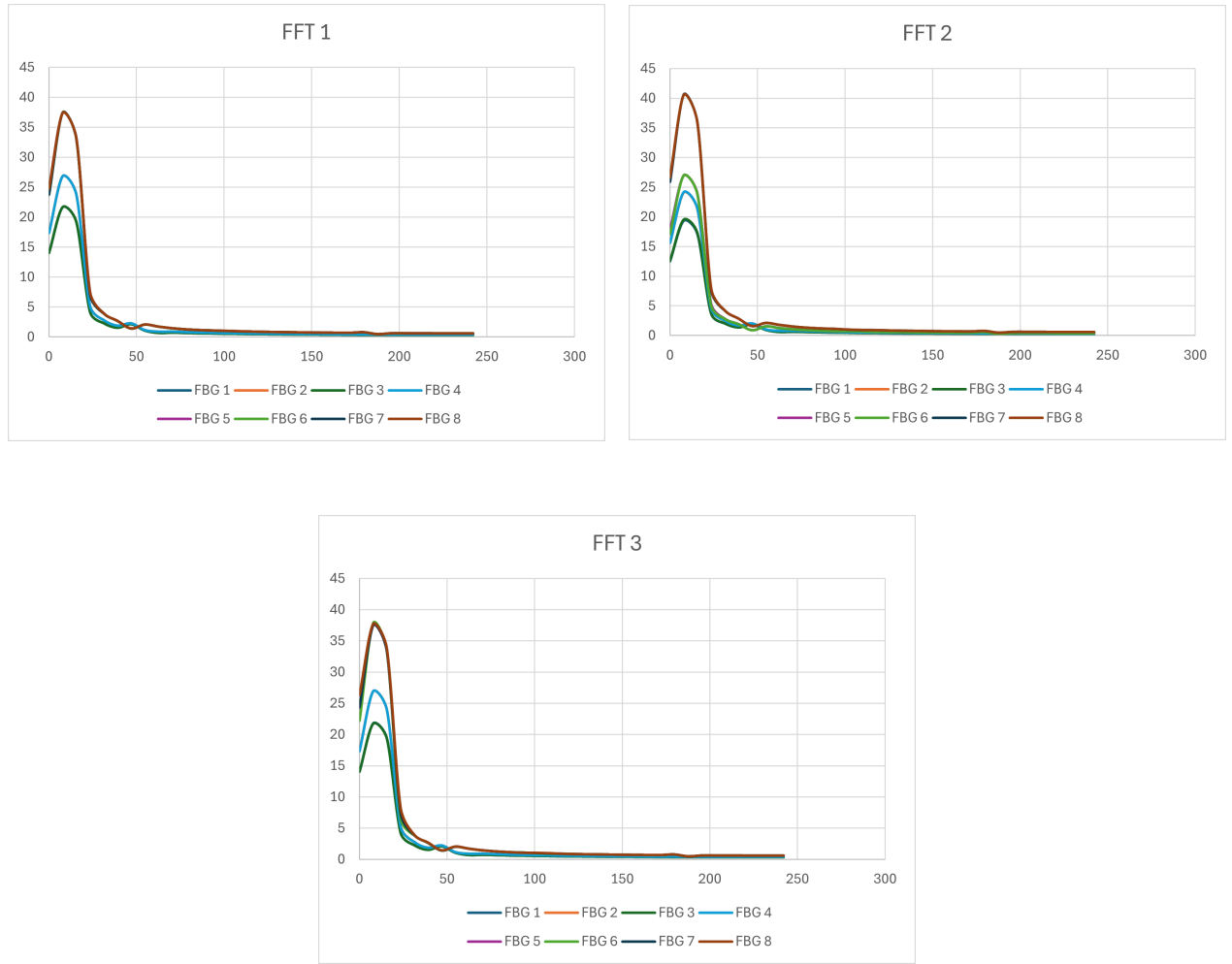
Signal Transforms

To complete the analysis, transforms of the strain signals in the interval $0 - 0.1\text{ s}$ were calculated.

Through *zero padding* the number of samples was brought to $N = 64$, with sampling frequency $f = 500\text{ Hz}$ and resolution $\delta f = 7.8125\text{ Hz}$.

The distribution of samples over a wider interval than the load period introduces *spectral leakage* phenomena. To mitigate this effect, a Hann window was applied:

$$w(n) = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi n}{N} \right) \right]$$



The results show two main peaks, corresponding to the characteristic frequency of the load (10 Hz) and the first natural frequency of the structure (approximately 50 Hz), in agreement with the modal analysis.

7.1 Generalization of the Solution

The symmetries observed in the three cases allow formulation of more general considerations on the inverse problem.

In particular, the motion of the source tends to:

- maximize the difference in strain between sensors aligned with the direction of propagation;
- minimize the difference between sensors orthogonal to this direction.

These observations hold under the hypothesis of linear regime and response dominated by the first flexural mode. In the presence of excitations at higher frequencies, behavior could become significantly more complex.

Once the load parameters have been identified qualitatively, this information can be used as a basis for subsequent quantitative reconstruction using inverse methods.

8 Least Squares Method

The inverse problem of identifying the mobile load is formulated as a least squares estimation problem. Problems of this type fall within the class of inverse problems and are extensively studied in the literature of parametric identification and model estimation from experimental observations [?, ?].

Indicating with $\boldsymbol{\varepsilon}_{\text{meas},i} \in R^N$ the vector of measurements acquired by sensor i and with $\boldsymbol{\varepsilon}_{\text{th},i}$ the corresponding theoretical response, the cost functional is defined as:

$$J(p_{\max}, x_0, y_0, u_0, v_0) = \sum_{i=1}^8 \|\boldsymbol{\varepsilon}_{\text{meas},i} - \boldsymbol{\varepsilon}_{\text{th},i}(p_{\max}, x_0, y_0, u_0, v_0)\|_2^2.$$

The model adopted has a *separable parameters* structure, in which the scalar parameter $p_{\max}$ appears linearly, while the kinematic parameters (x_0, y_0, u_0, v_0) enter nonlinearly. Since the dependence on $p_{\max}$ is linear, it is possible to separate the estimation of the load intensity from that of the kinematic parameters. This approach, known as variable separation in nonlinear least squares problems, was formalized by Golub and Pereyra [?].

In particular, it is assumed:

$$\boldsymbol{\varepsilon}_{\text{th},i} = p_{\max} \mathbf{s}_i(x_0, y_0, u_0, v_0),$$

where $\mathbf{s}_i$ represents the model response normalized with respect to load intensity.

Minimization with respect to $p_{\max}$, for fixed kinematic parameters, leads to the closed-form solution:

$$\hat{p}_{\max} = \frac{\sum_{i=1}^8 \boldsymbol{\varepsilon}_{\text{meas},i} \cdot \mathbf{s}_i}{\sum_{i=1}^8 \mathbf{s}_i \cdot \mathbf{s}_i},$$

which represents the least squares solution of the associated linear problem.

Substituting $\hat{p}_{\max}$ in the functional, the reduced problem is obtained:

$$J_{\text{rid}}(x_0, y_0, u_0, v_0) = \sum_{i=1}^8 \left(\boldsymbol{\varepsilon}_{\text{meas},i} \cdot \boldsymbol{\varepsilon}_{\text{meas},i} - \frac{(\boldsymbol{\varepsilon}_{\text{meas},i} \cdot \mathbf{s}_i)^2}{\mathbf{s}_i \cdot \mathbf{s}_i} \right),$$

which depends exclusively on the kinematic parameters.

The procedure adopted therefore consists of finding the parameters (x_0, y_0, u_0, v_0) that minimize J_{rid} . The resulting problem is a low-dimensional nonlinear optimization problem, treatable with standard numerical optimization techniques [?].

In the current implementation, minimization is performed using a multigrid strategy on a four-dimensional grid of kinematic parameters. Preliminary results show good reconstruction capacity of known kinematic conditions, although a study is still underway to improve robustness with respect to the presence of local minima.

9 Conclusions and Future Work

The work presented has addressed the problem of identifying moving dynamic loads acting on composite material structures from distributed strain measurements. The case study considered consists of a laminated panel equipped with optical Fiber Bragg Grating (FBG) sensors capable of providing time series of strain at discrete points on the surface.

In the first part of the work, a simplified analytical model of the structure was developed, based on Kirchhoff-Love theory for anisotropic laminated plates and Classical Laminate Theory for determining the flexural stiffness of the laminate. The dynamic response was obtained through an energy formulation and a Rayleigh-Ritz modal discretization, which allowed estimation of the system's natural frequencies.

Subsequently, a finite element numerical model of the real structure was constructed, including the simplified geometry of the region of interest, the definition of the laminate stack-up, and integration of sensors in the model. Modal analysis made it possible to identify the frequency band dynamically relevant to the excitation, while transient analysis made it possible to reconstruct the time series of strain at the sensor points. A convergence study was also conducted to verify the influence of spatial discretization, time step, and numerical damping on the results.

Analysis of the deformation signals showed how comparison between sensors symmetric and antisymmetric with respect to the structure geometry allows identification of characteristic patterns associated with the direction of propagation and possible eccentricity of the mobile source. Based on these qualitative observations, the inverse problem of load identification was formulated as a least squares estimation problem. By exploiting the separable parameter structure of the model, it was possible to analytically isolate the estimation of load intensity and reduce the problem to nonlinear minimization in only the kinematic parameters of the source.

9.1 Future Work

The work carried out represents a first step toward the development of systematic methods for identifying moving loads from distributed strain measurements. Among future developments, rigorous mathematical formulation of the inverse problem is planned first, with the objective of identifying theoretical conditions of invertibility of the measurement system as a function of the number of sensors, their spatial arrangement, and orientation.

In this direction, mathematical modeling will be further developed starting from the Rayleigh-Ritz analytical model, through numerical integration of the equations of motion in the modal domain. In particular, it is planned to exploit modal projection to reconstruct the dynamic response of the structure, allowing direct comparison with results obtained from the finite element model and better integration between analytical and numerical approaches.

Further developments will concern analysis of method robustness in the presence of measurement noise and possible introduction of regularization techniques to improve reconstruction stability. From the experimental perspective, design of a dedicated setup is planned to validate the method, with particular attention to controlled generation of underwater waves.

Finally, more realistic load models will be considered, capable of representing with greater accuracy the laws of pressure wave propagation in real applications. The behavior of the structure in the nonlinear regime will also be analyzed, which could manifest in the presence

of high-intensity waves and lead to triggering of instability phenomena, such as buckling effects or nonlinear coupling between vibration modes.